

Building a Better Wood Furnace

Wood has been burned as fuel for thousands of years, and over this time the art of wood burning has been refined from outdoor fires to highly advanced wood furnaces and boilers, or hydronic furnaces. Hydronic furnace, a term adopted by the American Society for Testing and Materials (ASTM), describes what was formerly known as a wood boiler. Wood-fired hydronic furnaces create hot water, but not steam. In the past, outdoor wood boilers have given wood burning a bad reputation, since they produce a lot of smoke, clog chimneys with creosote, are highly polluting, and require frequent reloading and cleaning. The designers of a new hydronic furnace, the Greenwood, have attempted to address all of these problems.

When a fire is kindled in a regular wood furnace, the temperature quickly rises to around 500°F. At this temperature, the wood begins to break down chemically and throw off volatile gases. These gases catch fire and accelerate the rise in temperature to about 1,100°F, at which point the wood solids begin to burn. If the temperature remains at or above 1,100°F, the fire consumes all of the wood gases and solid materials, and the fire is clean burning. If not, the combustion is incomplete, which means that unburned gases and particulates are vented by the furnace. Some of this effluvia escapes out the chimney as smoke; the rest cools down and remains in the chimney in the form of creosote. Any fuel that contains carbon (which includes coal, crude oil, and natural gas, as well as organic substances such as wood and straw) produces CO₂ when it is burned, which contributes to environmental degradation and climate change.

Most wood-burning furnaces and wood boilers on the market today are unable to sustain a temperature of 1,100°F or higher. They are built with a firebox made of steel, which is surrounded by a jacket of water. This water jacket serves two purposes: It transfers heat from the firebox to the home heating system, and it cools the steel firebox and keeps it from melting. The problem

is that by keeping the firebox cool, the water jacket also cools the fire and prevents it from burning at temperatures needed for complete combustion. That's why these units produce irritating smoke and potentially dangerous creosote.



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The Nickell family of Anchorage, Alaska, say that their Greenwood performs flawlessly.

New Design, New Results

Greenwood's hydronic wood furnace is built in an entirely different way from other wood-burning furnaces and boilers. The Greenwood's firebox is made of superduty ceramic refractory, cast 4–6 inches thick, and surrounded by outer layers of insulation designed to keep the heat in. The natural draft system pulls air into the furnace, which creates a roaring fire with sustained temperatures of 1,800°F–2,000°F. Heat from the fire is captured by a water tube heat exchanger located above the firebox in the path of the escaping superheated gases. The furnace extracts heat from these escaping gases, not from the fire below. The firebox itself is warm to the touch.

This innovative design, which has been in operation for over 20 years, enables the Greenwood furnace to burn

cleanly and operate at a very high level of efficiency. By the time the escaping gases leave the furnace, they have cooled to around 300°F. The 1,700°F difference in temperature between the firebox and the vent represents the amount of energy captured to heat the home. The Greenwood hydronic furnace delivers up to 87% of the wood's thermal energy to the home heating system. When the Greenwood is combined with a radiant or forced-air heating system, it can substantially reduce winter heating bills. If a Greenwood furnace is properly installed and there is adequate air flow into the unit and adequate draft from the flue, it will generally burn without visible smoke. During a cold start-up, or if old or wet wood is used, visible smoke may be produced.

Founded in 2005, Bellevue, Washington-based Greenwood Technologies manufactures the Greenwood in three different models, Model 100 (100,000 Btuhs); Model 200 (200,000 Btuhs); and Model 300 (300,000 Btuhs). The company was founded by Tom Eckmann and Dave Barber, who led a small team of investors to acquire the product rights from a small manufacturer in Montana. The Montana company had developed the initial product during the previous 20 years and had marketed it regionally. Eckmann and Barber brought in a team of seasoned and successful entrepreneurs who took the product through a substantial upgrade in 2005. The Greenwood furnace is as safe to operate as a water heater. It meets both strict U.S. AN SI/UL-391 and Canadian CSA B366.1 standards for indoor operation.

The Greenwood in Action

The hot water that the Greenwood furnace produces can be used for in-floor and baseboard hydronic radiant heating for a home, garage, and out-buildings. It can be used with forced-air heating, using a water-to-air heat exchanger that fits into existing ductwork. It can be used for domestic hot water (as a sole source or in conjunction with another water heater); or for heating a swimming pool.

In the Greenwood, the length of the burn depends on the type and quality of the wood being burned and on the demand for heat, which varies with the inside and outside temperature, the size of the home, the quality of the insulation, and other factors. As a general rule, however, the Greenwood firebox holds enough wood to burn for eight to ten hours during normal winter conditions and longer during milder weather.

The Greenwood furnace controls the rate of burn by controlling the amount of oxygen entering the firebox. When sufficient heat has been reached, the thermostat closes a damper over the air intake manifold, shutting off the flow of oxygen to the fire and stopping the burn. The 4–6-inch refractory walls, however, have absorbed a tremendous amount of radiant heat, which keeps the firebox hot. When additional heat is needed, the damper opens, and as long as the firebox temperature remains above 500°F, the furnace will automatically refire.

Logs are loaded into the firebox and ignited with paper and kindling. As the fire grows, fresh air is drawn through the air intake manifold, fanning the flames in the ceramic firebox. The burning wood gases reach 2,000°F before flowing out of the firebox and down the flame path toward the exhaust vent. As the superheated exhaust gases move toward the vent, their energy passes to fluid flowing through an internal heat exchanger. This heat transfer fluid reaches 180°F before circulating to an external heat exchanger mounted on the back of the furnace.

Aquastats (water thermostats) control the operation of the furnace by monitoring the temperature of the heat transfer fluid and regulating a damper on the air intake manifold. When the air in the house reaches the desired temperature, the damper closes, shutting off the flow of fresh air and extinguishing the fire. When more heat is needed, the damper opens and the furnace refires. Heat stored in the refractory walls of the firebox supports automatic refiring for up to two days.

Real-Life Results

Steve and Marie Nickell, who live in Anchorage, Alaska, had relied on a Burnham boiler that used oil before

they bought the Greenwood Technologies Model 100. On average, they had used 2,000 gallons of #2 stove oil per year to heat their home. With the Greenwood, this is no longer the case.

“Our stove has been in operation since late January,” Steve Nickell says, “and we use about 2 cords of wood per month in the coldest weather. The amount of ash is amazing! I scoop out approximately 3 gallons of ash once a month. The Nickells say that they worry much less about using the Greenwood than they did when they relied solely on the oil stove. “The oil stove would fail at least once a year when the electrical parts to the fuel gun and spark mechanism would fail in the middle of the night,” says Nickell.

“Our greatest surprise with the Greenwood Technologies furnace is its ability to heat our home effectively without using oil. It effectively and efficiently burns the wood we feed it, which is everything from punky, rotten wet wood to green moisture-laden birch and spruce. I plumbed the system into the oil-fired system myself, and

despite some mistakes I made, the furnace performs flawlessly.”

“I highly recommend Greenwood Technologies hydronic furnaces to everyone who can burn wood,” says Nickell. “It gives us satisfaction to wean ourselves from burning fossil fuel. Wood is a renewable resource, and we get great pleasure going into the forest to harvest deadfall wood.”



—Elka Karl

Elka Karl is an associate editor with Home Energy magazine.

FOR MORE INFORMATION:

The Greenwood Model 100 costs \$7,000. It can currently be purchased in every U.S. state except the state of Washington. It can also be purchased in many provinces in Canada. For more information, go to www.greenwoodfurnace.com.